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ChatGPT-4o as a Tool for Patient Education in Plastic Surgery

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Background: Artificial intelligence (AI), particularly large language models (LLMs), has demonstrated potential to improve patient communication by delivering accurate, readable, and empathetic medical information. In plastic surgery—a specialty where preoperative counseling is essential—patients often seek online information to understand procedures, risks, and recovery. However, limited research exists on ChatGPT-4o's utility in this context.

Methods: This prospective qualitative study evaluated ChatGPT-4o's responses to 25 standardized patient questions across 5 plastic surgery procedures: rhinoplasty, breast augmentation, abdominoplasty, blepharoplasty, and rhytidectomy. Each procedure was queried with 5 common preoperative questions covering indications, alternatives, risks, surgical steps, and recovery. Responses were reviewed independently by a board-certified plastic surgeon and 2 researchers for accuracy, completeness, and appropriateness. Strengths, weaknesses, omissions, and potentially unsafe guidance were identified and summarized.

Results: ChatGPT-4o provided generally accurate, well-structured, and patient-friendly answers across all procedures, with no unsafe recommendations. Strengths included clear explanations of surgical rationale, common risks, general procedural steps, and recovery expectations. The model promoted safety and professional consultation. However, notable limitations included a lack of procedural nuance, omission of less common but clinically important risks, failure to tailor guidance to individual variables, and incomplete recovery or post-operative care details.

Conclusions: ChatGPT-4o offers significant promise as a supplementary patient education tool in plastic surgery. Its ability to deliver coherent, empathetic, and accessible responses may help bridge health literacy gaps. However, it should not replace detailed, individualized surgeon-patient discussions. Further refinement and real-world validation are needed to enhance its clinical reliability and integration into patient care.

Key Words: Artificial intelligence, ChatGPT, patient education, plastic surgery

Artificial intelligence (AI) tools are rapidly growing in sophistication and utility for medical practice. AI assistance has shown superiority to provider-only models, including the detection of colorectal neoplasia and the interpretation of bronchoscopies.[1,2] While it is thought that surgical subspecialties are more insulated from AI intervention, AI has shown potential in risk stratification and surgical training.[3] AI and its role in patient communication is an emerging topic as AI's strength lies in its ability to communicate comprehensible, concise, and accurate medical information. However, there have been incidents where large language models (LLMs) have been shown to produce factually incorrect responses. These incorrect responses remain a significant barrier for AI being implemented as a useful tool in surgical training and education.[4]

Artificial intelligence chatbots may significantly improve patient comprehension by simplifying complex medical information and bridging health literacy gaps. Studies have shown that before surgery, patients are most interested in understanding the operative steps, risks and benefits of alternatives, long-term outcomes, and the recovery process. Surgeons, while typically focused on providing detailed information about patients' conditions and treatment options, have been shown to inadequately explore patients' emotional concerns.[5] Surgeons, while typically focused on providing detailed information about patients' conditions and treatment options, have been shown to inadequately explore patients' emotional concerns. This reveals a key communication gap: although both patients and surgeons may discuss treatment and procedural details, surgeons often do so from a clinical perspective, whereas patients seek reassurance and emotional clarity as well.[6] LLMs may help bridge this communication gap, as they have been shown to produce responses to patients' questions that are significantly more empathetic than those provided by surgeons.[7] Furthermore, a study found that the average readability of informed consent forms provided to plastic surgery patients exceeds the 12th-grade level. In contrast, the average American reads at a 6th to 8th-grade level. This indicates a significant communication barrier that AI tools could help address.[8]

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Given the increasing public use of ChatGPT for health information, evaluating ChatGPT-4o's effectiveness in addressing patient queries in plastic surgery is crucial. Patients often seek online information about procedures, thereby illustrating the importance of accurate, detailed, and realistic AI-generated responses.

Limited research exists on ChatGPT-4o's performance in plastic surgery patient education. This study assesses ChatGPT-4o's responses to standardized questions about rhinoplasty, breast augmentation, abdominoplasty, blepharoplasty, and rhytidectomy. We evaluated accuracy, clinical relevance, and completeness against surgical standards. Our goal is to highlight strengths, limitations, and implications for integrating AI into patient education.

METHODS

We conducted a prospective evaluation of ChatGPT-4o responses to standardized preoperative patient questions related to 5 commonly performed elective plastic surgery procedures—rhinoplasty, breast augmentation, abdominoplasty, blepharoplasty, and facelift. These information domains include rationale for surgery, alternatives, potential complications, procedural details, and postoperative expectations. For each procedure, we formulated 5 key questions as outlined by the American Society of Plastic Surgeons (ASPS). The questions were: (1) "How do I know if I need procedure?", (2) "What are the treatment alternatives to procedure?", (3) "What are the risks of procedure?", (4) "How is a procedure performed?", and (5) "What is the recovery process for procedure?".

Each question was input into the OpenAI ChatGPT-4o interface (accessible via chat.openai.com) for each of the 5 procedures, generating a total of 25 question–answer pairs. The model was used in its default mode without any plug-ins or internet access to simulate the most general experience. All queries were conducted on February 25, 2025, and were void of any identifying patient information. All questions were generic; therefore, this study did not involve human subjects and was exempt from institutional review board oversight.

A board-certified plastic surgeon and 2 study members reviewed each of the LLM's responses. For each question response, they evaluated accuracy, comprehensiveness, and appropriateness. Each reviewer identified notable strengths and weaknesses in the LLM output. We specifically noted if ChatGPT provided any inappropriate or unsafe recommendations. This was defined as advice contradicting standard medical guidelines or posing potential harm if followed. We also checked for major omissions, such as failing to mention a commonly known critical risk or step, and major inaccuracies.

Qualitative findings were then synthesized for each procedure across the 5 question categories. We tabulated the results, listing the positive aspects and negative aspects of ChatGPT's answers for each question category per procedure. Results are presented in summary tables. Narrative comparisons are made to known standard information from the medical literature and clinical practice.

RESULTS

Rhinoplasty

ChatGPT-4o accurately identified both cosmetic and functional indications for rhinoplasty, noting its role in correcting structural issues like deviated septum and nasal valve collapse that impair breathing. It also cited nasal asymmetry, dorsal humps, nostril size, and tip shape as common cosmetic concerns. Another strength is its recommendation of seeking professional evaluation from a board-certified plastic surgeon to evaluate candidacy. Specifically, ChatGPT-4o's procedural explanation aligns with how rhinoplasty typically involves general anesthesia and incisions either within the nostrils (closed approach) or along the columella (open approach). The AI correctly mentions that the surgeon reshapes bone and cartilage to achieve the desired aesthetic or functional result. However, it simplifies the technique by omitting procedural nuances such as the use of autologous cartilage grafts from the septum, ear (conchal), or rib (costal) for structural support.

ChatGPT-4o lists general surgical complications like bleeding, infection, scarring, and anesthesia risks and notes dissatisfaction with cosmetic outcomes, which may require revision surgery. However, it does not mention several less common but clinically significant complications such as septal perforation, hematoma, skin necrosis, prolonged edema, or persistent nasal obstruction.

The LLM noted initial swelling and bruising around the eyes and nose during the first 1 to 2 weeks, along with recommendations to avoid strenuous activity and keep the head elevated. It also correctly indicates that final results may take several months to fully appear. However, it overlooked scheduling follow-up visits for nasal splint and dressing removal—typically done within the first week—and monitoring for complications like infection or hematoma. The model also underrepresents variability in individual recovery, which can depend on surgical complexity, patient anatomy, and healing response.

Table 1, Supplemental Digital Content 1, http://links.lww.com/SCS/I741, summarizes the identified strengths (+) and weaknesses (−) in ChatGPT's responses for each question category for Rhinoplasty.

Breast Augmentation

ChatGPT-4o accurately cited increased breast fullness and projection and enhanced self-image as reasons for the procedure, but overlooked reconstruction after mastectomy, a common medical indication. The LLM accurately lists general and implant-specific risks—such as infection, capsular contracture, implant rupture, and the need for revision surgeries—and provides a basic explanation of surgical options, including subpectoral and prepectoral implant placement with either silicone or saline implants. However, ChatGPT-4o omits several critical points. When discussing alternatives like fat grafting, it does not address limitations such as volume unpredictability or the need for multiple sessions. It also excludes changes in nipple or breast sensation, Breast Implant-Associated Anaplastic Large Cell Lymphoma (BIA-ALCL), and interference with mammography. Procedurally, the model lacks depth, offering no insight into how implant type or placement is chosen, and it skips over key surgical elements like operative duration, outpatient nature, and the use of drains. Similarly, its recovery explanation does not mention follow-up appointments or scar care, and it underrepresents variability in long-term outcomes.

Table 2, Supplemental Digital Content 1, http://links.lww.com/SCS/I741, displays a review of ChatGPT-4o's responses to breast augmentation questions asked before surgery.

Abdominoplasty

ChatGPT-4o accurately stated that an abdominoplasty is intended for individuals with excess skin or weakened abdominal muscles unresponsive to diet and exercise. More significantly, it appropriately distinguished abdominoplasty from weight-loss surgery. It emphasized that abdominoplasties are body-contouring procedures and introduced basic alternatives

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like exercise and liposuction. The AI also identified key surgical risks such as infection, hematoma, scarring, blood clots, and seroma. In addition, it noted that smokers and patients with health conditions may have a higher risk of poor wound healing. However, several important clinical nuances were missing. The model did not clearly state that abdominoplasty is an elective procedure driven by personal goals. Nor did it highlight the importance of weight stability and completion of childbearing for optimal long-term results. In its discussion of alternatives, it failed to mention newer minimally invasive treatments (eg, radiofrequency or laser-based skin tightening) and did not clarify that partial procedures like a "mini-tummy tuck" may be viable in select cases. While the procedural steps were correctly outlined in general terms, the explanation lacked technical depth and omitted common details like muscle plication or the possible use of drains. Recovery information addressed activity restrictions and use of compression garments, but failed to note the need for routine follow-up, scar care, or how to recognize early signs of complications such as infection or seroma.

Preoperative patient questions about abdominoplasty and ChatGPT-4o's corresponding responses are reviewed in Table 3, Supplemental Digital Content 1, http://links.lww.com/SCS/I741.

Blepharoplasty

ChatGPT-4o provided a general overview of blepharoplasty, correctly identifying both functional indications, such as visual field obstruction due to dermatochalasis, and cosmetic concerns like eyelid puffiness or wrinkling for the surgery. It appropriately advised consultation with a qualified plastic surgeon or ophthalmologist and outlined standard non-surgical alternatives, such as Botox. The LLM did not, however, specify that Botox is mainly effective for dynamic wrinkles and not redundant skin or ptosis. AI listed common surgical risks such as bleeding, scarring, dry eyes, and even rare complications like vision loss. It also noted surgical techniques for both upper and lower eyelid blepharoplasty, including incision placement in the supratarsal crease and subciliary or transconjunctival approaches. However, the AI omitted critical adjunctive procedures like canthopexy or canthoplasty, often required in lower lid surgery to maintain lid position and prevent ectropion.

Lastly, ChatGPT 4.0 correctly advised rest, head elevation, and avoidance of strenuous activity, but it failed to mention standard care practices like the use of cold compresses during the first 24 to 48 hours to reduce swelling and bruising, prescribed lubricating eye drops or ointments, and restrictions on contact lens use and eye makeup.

A review of ChatGPT-4o responses to preoperative patient concerns before blepharoplasty is shown in Table 4, Supplemental Digital Content 1, http://links.lww.com/SCS/I741.

Rhytidectomy

ChatGPT-4o accurately described rhytidectomy as an elective cosmetic procedure aimed at reducing signs of facial aging, and it appropriately emphasized that ideal candidates are healthy, non-smoking individuals with realistic expectations. It also recognized non-surgical alternatives, including Botox, dermal fillers, and energy-based skin-tightening devices, and correctly conveyed their limited, temporary results. The model effectively conveyed core surgical risks—hematoma, infection, nerve injury, and scarring.

In terms of procedure, ChatGPT 4.0 did not differentiate between deep plane facelift, SMAS (superficial musculoaponeurotic system) plication or imbrication, mini-lift, and mid-facelift. These distinctions matter clinically and depend on the patient's anatomy, aging pattern, and desired results. The model also neglected to mention that rhytidectomy is often combined with neck lift, blepharoplasty, or fat grafting.

Regarding recovery, while ChatGPT-4o noted swelling, bruising, head elevation, and compression garments, it left out the common use of postoperative drains and failed to provide specific timelines for resuming daily activities, such as returning to work or exercise. While the model listed standard risks like hematoma, infection, nerve injury, and scarring, it did not include alopecia along incision lines or the fact that hematomas are the most common complication and may require urgent reoperation.

Table 5, Supplemental Digital Content 1, http://links.lww.com/SCS/I741, summarizes ChatGPT-4o's strengths and limitations in addressing facelift-related patient questions.

DISCUSSION

Across all 5 procedures examined—rhinoplasty, breast augmentation, abdominoplasty, blepharoplasty, and rhytidectomy—ChatGPT-4o consistently demonstrated strengths in delivering accurate, patient-friendly overviews that captured core indications, general procedural steps, and major surgical risks. It was especially effective in identifying common motivations for these primarily elective surgeries and in recommending consultation with a board-certified plastic surgeon. It also performed well in highlighting typical recovery expectations and advising activity restriction. However, the model repeatedly fell short in conveying nuanced clinical details essential to surgical decision-making. It often omitted key anatomic or candidacy considerations—such as skin elasticity, bone structure, or age factors—and failed to distinguish between procedural variations (eg, deep plane versus mini facelift, open versus closed rhinoplasty). Its discussions of alternatives were also limited, frequently missing surgical options like brow lifts or neck procedures and underexplaining the limitations of temporary treatments like fillers or Botox. Risk explanations, while generally accurate, lacked depth—excluding serious but rare complications (eg, BIA-ALCL, vision loss, alopecia along incision lines) and the management implications of common issues like hematoma. Recovery guidance similarly omitted standard practices like the use of drains, eye drops, scar care, and follow-up visits. Overall, ChatGPT-4o provided reliable baseline education but lacked the clinical specificity and individualized detail needed for comprehensive patient counseling.

ChatGPT-4o demonstrates a significant improvement over prior iterations of the LLM. Notably, ChatGPT-4o answered ~74.4% of plastic surgery board exam questions correctly, a substantial leap from the 55.5% accuracy rate achieved by its predecessor, ChatGPT-3.5 ($P < 0.001$).[9] Similarly, this marked improvement has been seen on the USMLE Step 2 exam and Medical College Admission Test (MCAT).[10,11] In another study, ChatGPT-4o also outperformed its predecessor in answering common patient questions about osteoarthritis, with orthopedic subspecialists rating its responses as more aligned with expert reasoning than those of ChatGPT-3.5. However, despite its improved coherence and completeness, ChatGPT-4o's responses still lacked the depth and contextual nuance of human-generated content, echoing the limitations we observed in our study and highlighting an important area for improvement.[12]

ChatGPT-4o may be beneficial in patient communication as it helps address discrepancies in health literacy. Low health literacy is associated with over $100 billion annually in increased hospitalizations and is a stronger predictor of a person's

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health than age, income, employment status, education level, and race.[13,14] As the majority of Americans have a reading level of approximately the eighth grade, the American Medical Association (AMA) recommends that patient education materials be written at or below a sixth-grade reading level to ensure accessibility for the average U.S. adult.[15,16] However, numerous studies have found that existing patient education materials often exceed this recommended level, potentially hindering patient understanding.[8,17,18] A study aimed to evaluate ChatGPT-4o's effectiveness in enhancing the readability of patient-focused materials in hand surgery. Findings revealed that ChatGPT-4o improved readability across all assessed formulas, achieving a mean sixth-grade level according to the Flesch-Kincaid Grade Level and Simple Measure of Gobbledygook calculations. This would align with the American Medical Association's recommendations for patient education materials.[16] This suggests ChatGPT-4o has the potential to serve as a valuable adjunctive tool for generating more accessible content that empowers patients to make informed health decisions.

This study has several important limitations that should be acknowledged. First, this study is qualitative in nature and did not employ validated assessment tools, quantifiable scoring systems, or Likert-scale metrics to evaluate ChatGPT-4o's responses. As such, the analysis inherently lacks numerical rigor or statistical generalizability. While expert review was used to assess response quality, the absence of standardized benchmarks limits reproducibility and comparative evaluation across studies or LLMs. Second, the findings represent a snapshot of ChatGPT-4o's capabilities as of February 2025; given the model's continual updates and retraining on larger datasets, its performance is likely to evolve, rendering this evaluation temporally constrained. Third, the model was assessed using a fixed set of 25 standardized questions across only 5 plastic surgery procedures. Though these questions reflect common preoperative concerns, they may not encompass the breadth, variability, or complexity of patient inquiries encountered in real-world clinical interactions. Lastly, the study was conducted in a controlled, static environment without dynamic patient dialogue, thereby limiting assessment of the LLM's capacity to personalize responses, manage follow-up questions, or adjust communication tone. These limitations underscore the need for more comprehensive, mixed-methods research that includes real-time patient interactions, outcome metrics, and broader procedural coverage to fully characterize the role of LLMs in surgical education and communication.

CONCLUSIONS

ChatGPT-4o demonstrates meaningful potential as an adjunctive tool for plastic surgery patient education, offering accurate, coherent, and comprehensible responses to commonly asked preoperative questions. Across 5 core procedures, its outputs reflected a strong command of foundational knowledge, with a tone that was both informative and safety-oriented. At the same time, the model lacked the depth, nuance, and individualized patient guidance; its ability to deliver accessible, empathetic content positions it as a valuable asset for enhancing patient comprehension—particularly in settings where health literacy is a barrier. However, consistent omissions of procedural specifics and context-sensitive factors underscore the need for cautious use. ChatGPT-4o should not be seen as a replacement for surgeon-patient dialogue but rather as a supplementary resource that may improve baseline understanding and engagement.

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Stakeholders in the Implementation of Artificial Intelligence in the Surgical Field: Challenges and Considerations Ahead

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